

Radical equations



1 Solve the following equations:

a $\sqrt{x} = \sqrt{13}$

b $\sqrt{h} = 5$

c $\sqrt{x} = \frac{3}{8}$

d $-3\sqrt{k} = -12$

2 Define a radical equation.

3 Is 4 a solution of the equation $\sqrt{x} = -2$?

4 Is 3 a solution of the equation $\sqrt{x-2} - 4 = \sqrt{x+6}$?

5 Is -2 a solution of the equation $\sqrt{x+18} = x+6$?

6 Solve the following equations:

a $\sqrt{y} - 6 = 1$

c $\sqrt{p} + 4 = 7$

e $\sqrt{x} - \frac{1}{4} = \frac{1}{3}$

g $5\sqrt{y} + 4 = 14$

i $2(\sqrt{b} - 3) = 4$

k $\frac{\sqrt{b} + 1}{2} = \frac{3}{5}$

b $5\sqrt{x} + 3 = 28$

d $\sqrt{y} + \frac{1}{5} = \frac{4}{5}$

f $\frac{\sqrt{x}}{-2} = -4$

h $-5\sqrt{c} - 2 = -17$

j $\frac{\sqrt{a} - 1}{2} = 1$

l $5(3\sqrt{a} + 8) = 70$

7 Solve the following equations:

a $\sqrt{x-5} = \sqrt{25}$

c $\sqrt{x-5} = 7$

e $\sqrt{9x^2 - 48x + 64} = x$

g $8\sqrt{x} - 4\sqrt{x} = 16$

i $\frac{\sqrt{x+9}}{2} - 8 = -2$

k $\sqrt{x^2 + 20} = 6$

b $\sqrt{2x-6} = 2$

d $\sqrt{x^2 - 8x + 16} = 7$

f $\frac{12}{\sqrt{x-2}} = 3$

h $\sqrt{3x-2} = 8$

j $\frac{7\sqrt{x}}{11} + \frac{2\sqrt{x}}{11} = 9$

- 8 Ray is given the equation $\sqrt{x+3} + 7 = 4$ to solve. His work is shown:



- a Does $x = 6$ satisfy the equation $\sqrt{x+3} + 7 = 4$?
- b Describe the error Ray made in his work.

1 $\sqrt{x+3} + 7 = 4$

2 $\sqrt{x+3} = -3$

3 $x + 3 = 9$

4 $x = 6$

- 9 Tom and Katrina are both given the equation $2\sqrt{x+71} - 10 = 8$ to solve. Both have made errors in their work below:

Tom's work

1 $2\sqrt{x+71} - 10 = 8$

2 $\sqrt{x+71} - 10 = 4$

3 $\sqrt{x+71} = 14$

4 $x + 71 = 196$

5 $x = 125$

Katrina's work

1 $2\sqrt{x+71} - 10 = 8$

2 $2\sqrt{x+71} = 18$

3 $\sqrt{x+71} = 9$

4 $x + 71 = 3$

5 $x = -68$

- a Describe Tom's error.
- b Describe Katrina's error.
- c Solve the equation $2\sqrt{x+71} - 10 = 8$.
- 10 Consider the equation $\sqrt{x+3} + 10 = 0$. Without solving, explain how we can see that the equation has no real solutions.
- 11 Consider the equation $10x = 1 + 3\sqrt{x}$.
- a Solve the equation for x .
- b Which value of x is an extraneous solution?
- 12 Consider the equation $\sqrt{x-5} = \sqrt{x} - 1$.
- a Solve the equation for x .
- b Is this value of x an extraneous solution?



14 Solve the following equations:

a $\sqrt[3]{x+7} = 4$

b $\sqrt[3]{3x-4} = 2$

c $\sqrt[4]{x+2} + 4 = 1$

15 Solve the following equations:

a $x^{\frac{3}{2}} = 27$

b $x^{\frac{2}{7}} = 4$

c $4x^{\frac{1}{4}} - 3 = 9$

d $4x^{\frac{2}{7}} - 1 = 15$

Applications

16 A square pyramid with a height of 12 units and a volume of V cubic units has a base with a side length of $b = \sqrt{\frac{V}{4}}$ units.

a Find the side length of the base of a square pyramid for each of the volumes in the table. Round your answers to the nearest tenth of a unit.

V	40	400	4 000
b			

b Each volume in the table increased by a factor of 10. Did the corresponding side length of the base increase by a factor of 10?

17 An electron is constantly moving, and its position with respect to the fixed point ($x = 0$) can be modeled by the equation $x = -(t - 4)^2 + 4$, where x is its position after t seconds. After 4.5 seconds, it is at $x = 3.75$.

a Solve the equation for t .

b Determine the time t at which the electron is at the fixed position.

c Determine how long it takes the electron to get from $x = -5.1$ to $x = -6.3$. Round your answer to the nearest tenth of a second.

- 18 The Beaufort Wind Scale was devised by British Rear Admiral Sir Francis Beaufort, in 1805 based upon observations of the effects of the wind. Beaufort numbers, B , are determined by the equation $B = 1.69\sqrt{s + 4.45} - 3.49$, where s is the speed of the wind in mph, and B is rounded to the nearest integer from 0 to 12.



- a Using the table shown, find the Beaufort number of a speed of 40 mph.
- b Classify the force of wind at a speed of 40 mph.
- c In 1946, the scale was extended to accommodate strong hurricanes. A strong hurricane received a B value of exactly 15. Determine the value of s to the nearest whole number.
- d Any B values that round to 10 receive a Beaufort number of 10. Using technology, find an approximate range of wind speeds, to the nearest mile per hour, associated with a Beaufort number of 10.

Beaufort Wind Scale	
Beaufort number	Force of wind
0	Calm
1	Light air
2	Light breeze
3	Gentle breeze
4	Moderate breeze
5	Fresh breeze
6	Steady breeze
7	Moderate gale
8	Fresh gale
9	Strong gale
10	Whole gale
11	Storm
12	Hurricane